

Forefoot Problems in Sport

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Great caution must be exercised in dealing with forefoot problems, particularly in athletes. Although the outcomes are the same as in nonathletes, even a small decrease in performance may prove to be very significant for an athlete. Therefore with certain exceptions, forefoot surgery should be considered only when conservative management has failed and the athlete is no longer able to play through the pain/dysfunction.

TURF TOE

Metatarsophalangeal (MTP) joint injuries are commonly the result of hyperextension. They are increasingly common and can be characterized by a significant delay in return to sporting activities.¹⁻⁴ Turf toe is more likely to occur on third-generation artificial surfaces versus grass, and is more than 13 times more likely to occur during a game as opposed to training.⁵ Analysis of the National Collegiate Athletic Association (NCAA) Injury Surveillance System for Turf Toe in Intercollegiate Football found an overall incidence of 0.062 per 1000 athlete-exposures.⁵ In a series of 20 male cadaveric feet, the first MTP joint was dorsiflexed to varying degrees and subsequently dissected to identify objective evidence of injury to the sesamoid complex, including tear in the soft tissue or avulsion fracture (deemed clinically relevant). Using binomial regression, a hallux dorsiflexion angle of 78 degrees had a 50% risk of first MTP joint sprain.⁶

History

Players may report localized pain, swelling, and pain with range of motion (ROM) and ambulation; in addition, periarticular swelling and ecchymosis are typically present. Push-off is impaired with running, and it may be difficult to crouch with the MTP joint extended.^{7,8} Some patients, particularly elite footballers, may not present with an acute history and severe symptoms but may have an impending turf toe, so they should have early imaging.⁹

Physical Examination

Clinical evaluation should include evaluation of hallux alignment, including clinical MTP joint reduction, or the presence of an intrinsic-minus posture (i.e., MTP extension and interphalangeal flexion), suggesting disruption of the flexor hallucis brevis (FHB) tendon insertion. Palpation of plantar (including the individual sesamoids), medial, lateral, and dorsal structures for focal tenderness should be performed. Assessment of joint stability with respect to dorsoplantar and varus/valgus stress is

critical. Weakness in plantarflexion strength of the hallux may indicate loss of integrity of the plantar plate or the FHB.¹⁰ Point tenderness over the plantar aspect of the first MTP joint that worsens with passive extension of the toe is indicative of injury.

Imaging

Bilateral weight-bearing anteroposterior (AP) views can identify proximal migration of the sesamoids. A forced dorsiflexion lateral view assists in the diagnosis of failure of distal sesamoid migration and diastasis of a bipartite or fractured sesamoid. The normal diastasis of a dominant foot bi- or multipartite sesamoid is on average 0.79 mm; it has been suggested that the sesamoid interval of more than 2 mm on AP radiograph of the foot is abnormal, with no statistically significant difference between gender, age, laterality of sesamoid, or foot dominance.¹¹ Weight bearing and stress radiographs of both feet are useful to document relative proximal migration of the sesamoids on the injured foot.¹⁰ Magnetic resonance imaging (MRI) plays a crucial role in early and accurate diagnosis of structures involved in "turf toe" injuries.¹² Systematic assessment of the MRI can detail the integrity of the paired confluent metatarsosesamoid (MT-S) ligaments, paired sesamoid-phalangeal (SP) ligaments, the intersesamoid (IS) ligament, medial and lateral collateral ligaments, accessory collateral ligaments to the sesamoids, tendinous insertions and confluences with the plantar capsule, as well as the sesamoids, cartilage and alignment.¹³ It is important to focus MR imaging on the first MTP joint during the assessment of turf toe by using an extremity coil with the joint at its isocenter as opposed to imaging the entire foot or forefoot.¹³

MRI can be used to evaluate the presence and extent of capsular or plantar plate disruption¹⁴ and also osseous or articular damage in the presence of normal radiographs.¹⁵ Rodeo and coworkers¹⁶ have proposed a classification of first MTP joint injury, as shown in [Box 121.1](#).

Decision-Making Principles

Operative intervention is rarely necessary in acute grade I and II injuries.⁸ Management is usually centered on conservative methods because surgery can be associated with a restricted ROM. However, up to 25% to 50% of patients will have residual pain and limited dorsiflexion despite 6 months of rehabilitation¹⁷; some authors have therefore recommended surgical treatment of type III injuries.¹⁸ On average, ligamentous turf toe results in a mean 9.1 days lost from play, whereas sesamoid fracture

Abstract

This chapter discuss the principles and current literature for management of forefoot problems in sport. In particular it focuses on the management of turf toe, sesamoiditis, sesamoid dysfunction, metatarsalgia, and hallux rigidus. The video highlights present important concepts for managing these patients. Further reading and references are provided in the chapter text.

Keywords

hallux rigidus
metatarsalgia
sesamoiditis
sesamoid dysfunction
turf toe

BOX 121.1 Classification of First Metatarsophalangeal Joint Injuries

Grade 1: Acute Sprain of the First Metatarsophalangeal Joint Plantar Capsule

Localized tenderness, swelling, and pain with dorsiflexion
Normal radiographs
Conservative treatment

Grade 2: Acute Sprain of the First Metatarsophalangeal Joint With Significant Plantar Capsule Disruption

Ecchymosis, painful dorsiflexion, and loss of motion
Diastasis of a partite sesamoid or joint instability on radiographs
No degenerative first metatarsophalangeal joint changes
Conservative or surgical treatment

Grade 3: Chronic Symptoms Involving the First Metatarsophalangeal Joint Resulting From Previous Injury

Loss of motion
Degenerative joint disease, hallux rigidus, or malalignment on radiograph
Treatment is often surgical

BOX 121.2 Sesamoid Function

Protects the tendon of the flexor hallucis longus
Absorbs most of the weight bearing on the medial aspect of the forefoot
Increases the mechanical advantage of intrinsic musculature of the hallux
Elevates the first metatarsal head

results in a mean of 42 days lost.⁵ Some patients will be out of elite-level competition for almost 6 months.⁹

Treatment Options

Nonoperative

A recent algorithm for the management of turf toe was suggested by Hong et al. in 2016. Grade 1 injuries can be treated with taping of the toe in slight plantarflexion for 3 weeks followed by a carbon fiber orthotic for 6 to 8 weeks. Grade 2 injuries are treated with a walking boot and crutches for 2 to 4 weeks and then, again, with a carbon fiber turf toe orthotic for 6 to 8 weeks. It is suggested that grade 3 injuries need surgical repair. Surgery should also be considered for sesamoid retraction or fracture, traumatic hallux valgus, first MTP joint instability, intra-articular loose bodies, or failed conservative treatment.¹⁹ Ice, compression, and nonsteroidal anti-inflammatory drugs may be used.^{1,7,20,21} However, activity should be restricted in athletes with more severe injuries and significant discomfort. The patient can continue to participate in athletics if pain is minimal. A rigid forefoot insole^{1,7,20} to stiffen the shoe and taping of the toe to compress the joint and limit dorsiflexion motion all help reduce pain. For grade III injuries, immobilization and a period of restricted weight bearing is usually required for up to 8 weeks.

Return to sport depends on the severity of the injury, the resolution of symptoms, and the athlete's sport or position. Ideally, painless 60-degree dorsiflexion will be achieved in the first MTP joint before running or explosive activities are attempted. Serial examinations are important because deformity can progress with athletic activity.²¹

Operative

Less than 2% of cases require operative treatment.⁵ Operative treatment is necessary in more severe injuries with joint instability, retraction of the sesamoids, diastasis of a bipartite

sesamoid or sesamoid fracture, traumatic hallux valgus, or the presence of a loose body or chondral injury. Arthroscopy is helpful in evaluating the injury, but an open approach (plantar medial with extension across the MTP joint if required) is most helpful; however, care must be taken with the plantar-medial nerve. Repair of the plate distal to the sesamoid can be achieved with suture anchors into the base of the proximal phalanx.⁹ Care must be taken to avoid overtightening this repair. Despite open repair and direct visualization of the lateral sesamoid, it is recommended that intraoperative imaging may help the surgeon ensure that the plantar plate is tensioned adequately.⁹ Similar techniques can be used for diastasis of a bipartite sesamoid. Postoperatively, patients are often placed into cast, boot, or splint for 4 to 6 weeks, followed by a resistive ROM program and running at 18 weeks.¹⁰ Return to full participation in collegiate football can take 6 months.¹⁰

Postoperative Management

Postoperative management must balance soft tissue healing with early ROM of the MTP joint.¹⁸ Immobilization in a removable splint (a toe spica splint in 5 to 10 degrees of plantarflexion with slight varus) with restricted weight bearing continues for 4 weeks.²¹ Seven to 10 days after surgery, passive ROM is begun. At 4 weeks, weight bearing and active ROM are initiated in a walking boot. Wearing of modified footwear (i.e., a shoe with a stiff sole modified with a turf toe plate) is allowed at 8 weeks, with return to full sport at 3 to 4 months.

Results

Contrary to the common perception, evidence points to good outcomes when appropriate treatment is instigated. Most athletes with grade I and II injuries (53 of 56 in a series by Clanton et al.¹) returned to sport within 3 weeks with careful conservative management. Even with grade III injuries, Anderson found that seven of nine players returned to sport; two failed to resume playing because of pain and degeneration.¹⁸

Complications

After a turf toe injury, athletes can experience persistent pain with toe-off and stiffness of the first MTP joint (Table 121.1). Instability, deformity, and degeneration can occur if the injury fails to heal well.

SESAMOID DYSFUNCTION

The sesamoid complex of the first MTP joint plays a significant role in the function of the great toe (Box 121.2). Sesamoid dysfunction is uncommon; however, it can occur with arthritis,^{22–26}



Authors' Preferred Technique

Turf Toe

The patient is positioned supine with the heels at the end of the operating table. Dual incisions are recommended to repair a grade III turf toe (Fig. 121.1). A longitudinal medial incision is centered at the first metatarsophalangeal joint to access the medial sesamoid.

Care is taken during dissection to leave a cuff of joint capsule for closure at the conclusion of the case. The medial plantar digital nerve lies adjacent to the sesamoid and should be protected. A longitudinal plantar incision is commonly used for the lateral sesamoid. The lateral plantar digital nerve is at risk during this approach and must be identified and protected to prevent neuroma formation.

The plantar plate is then repaired from lateral to medial with use of nonabsorbable or slow-absorbing suture. Care must be taken to avoid damage to the flexor hallucis longus. In the absence of tissue along the proximal phalanx, use of a suture anchor to secure the plantar plate has been described. Fracture of both sesamoids should be treated with cerclage of the sesamoids to restore continuity of the flexor hallucis brevis and avoid a cock-up toe deformity. If one of the two sesamoids has severe articular damage, excision can be performed. However, in no case should both sesamoids be excised. Standard closure should be performed.



Fig. 121.1 Dual incisions are required to repair a turf toe. Each individual incision can also be used for an isolated sesamoidectomy. Care must be taken to protect the plantar digital nerve during dissection. (From Kadakia AR. Disorders of the hallux sesamoids. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

trauma,^{22,24,27–36} osteochondritis,^{29,37–39} infection,^{40–44} or sesamoiditis.^{23,38,39,45} Abnormalities and complaints regarding the sesamoids of the hallux are much more common in professional athletes than in others because of the stresses the first MTP joint is subjected to during athletic activities.^{46,47}

The medial sesamoid is more commonly injured than is the lateral sesamoid because it bears greater force during normal gait.⁴⁸ The plantar lateral and plantar medial digital nerves are located adjacent to the lateral and medial sesamoids. Impingement

of either one of these branches may be a source of pain in the area of the sesamoids. Osteoarthritis of the sesamoids has been reported^{22,23,25,37,49} in association with hallux valgus, hallux rigidus, and rheumatoid arthritis and as an isolated occurrence. When it is associated with a high-arched or cavus type of foot, a plantarflexed first ray may be the cause of the callus formation. When a sesamoid is involved, a more localized or concentric lesion is usually present.

History

The most frequent subjective symptoms are pain and discomfort in the toe-off phase of gait located directly plantar to the sesamoids.

Physical Examination

A cavus foot or gastrocnemius contracture increases forefoot loading and therefore should be specifically assessed. A plantar keratosis beneath either the tibial or fibular sesamoid may occasionally accompany a symptomatic sesamoid. Another method to support the diagnosis of sesamoiditis is the use of the passive axial compression test.⁵⁰ This is performed with the patient supine. After the sesamoids have been isolated, the hallux is maximally dorsiflexed at the MTP joint, which will cause distal migration of the sesamoids. The examiner's index finger is used to apply compression just proximal to the sesamoids. The patient's symptoms are reproduced with passive plantarflexion of the MTP joint as the sesamoids are compressed against the metatarsal head and phalanx.

Imaging

Often the most useful radiograph is the axial sesamoid view (Fig. 121.2). Fragmentation of a sesamoid in persons with osteochondritis may be seen on the axial radiograph. Radiographs are frequently normal despite subjective symptoms (sesamoiditis). MRI is one of the best methods for diagnosing disease of the sesamoids. Osteochondritis can be visualized very early, usually before any abnormality is seen on plain radiography. It is also useful in visualizing soft tissue abnormalities of the sesamoid mechanism.^{51,52}

Decision-Making Principles

Most sesamoid problems in the athlete can be treated effectively without surgery. If conservative treatment has been unsuccessful, several surgical options are available depending on the pathologic condition. When surgical intervention is being considered, a concomitant deformity such as a gastrocnemius contraction or cavus foot alignment must be addressed.

Treatment Options

Nonoperative

Activity and footwear modification, use of insoles, and treatment of the callosities are all helpful. In patients with sesamoid pain, a stiff insole or a rocker-bottom sole reduces metatarsal joint motion and can relieve pain with ambulation. The use of a dancer's pad, which is designed with a concavity for the sesamoids, is extremely useful in nearly all cases.

TABLE 121.1 Anatomic Classification of Turf Toe

ANDERSON CLASSIFICATION ^{127,128}		
Type of Injury	Grade	Description
Hyperextension (turf toe)	I	Stretching of plantar capsular ligamentous complex Localized tenderness, minimal swelling, minimal ecchymosis
	II	Partial tear of plantar capsular ligamentous complex Diffuse tenderness, moderate swelling, ecchymosis Restricted movement with pain
	III	Frank tear of the plantar capsular ligamentous complex Severe tenderness, marked swelling, and ecchymosis Limited movement with pain, plus vertical Lachman test Possible associated injuries: Medial/lateral injury Sesamoid fracture/bipartite diastasis Articular cartilage/subchondral bone bruise
Hyperflexion (sand toe)		Hyperflexion injury to hallux MTP or IP joint May involve injury to lesser MTP joints as well
Dislocation	I	Dislocation of the hallux with the sesamoids No disruption of the intersesamoid ligament Frequently irreducible
	IIA	Associated disruption of the intersesamoid ligament Usually reducible
	IIB	Associated transverse fracture of one or both sesamoids Usually reducible
	IIC	Complete disruption of intersesamoid ligament with fracture of one of the sesamoids Usually reducible

IP, Interphalangeal; MTP, metatarsophalangeal.

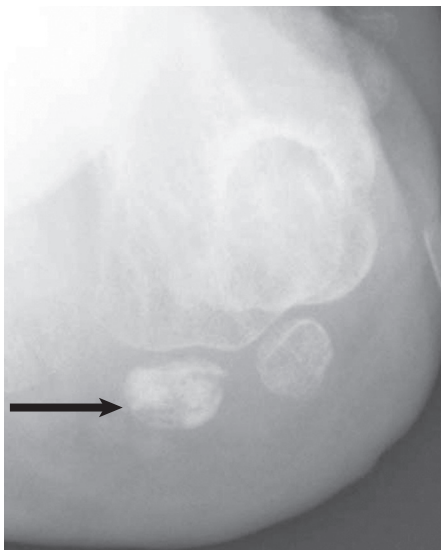


Fig. 121.2 Sesamoid views allow visualization of the metatarsal-sesamoid articulation. This view clearly demonstrates sclerosis (arrow), flattening, arthritis, and fracture. (From Kadakia AR. Disorders of the halluxal sesamoids. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

Operative

Surgical treatment includes a tibial or fibular sesamoidectomy as well as sesamoid shaving. Excision of a single sesamoid does not usually lead to deformity as long as careful operative technique is used.⁵² Removal of both tibial and fibular sesamoids is rarely

indicated because it predictably results in the development of a cock-up and intrinsic-minus deformity of the hallux. Arthrodesis of the MTP joint with excision of the sesamoid may alleviate arthritic symptoms and provides a stable medial buttress to the first ray but it will severely compromise function in an athlete. Although a sesamoidectomy is occasionally necessary for cases of intractable plantar keratosis (IPK), surgical shaving of the involved sesamoid may preserve function in the athlete and facilitate a more rapid recovery (Fig. 121.3).^{51,52} Sesamoid shaving or resection in the presence of a plantarflexed first ray is associated with a high rate of recurrence of the keratotic lesion.^{53,54}

 Authors' Preferred Technique
Sesamoid Dysfunction

Excision of the affected sesamoid with use of the previously described incisions is the treatment of choice except in the case of an intractable plantar keratosis. Care must be taken to avoid damage to the flexor hallucis brevis and flexor hallucis longus. The soft tissue defect must be repaired to avoid iatrogenic deformity.

Postoperative Management

After sesamoid excision, a soft compression dressing is used and the patient is allowed to ambulate in a postoperative shoe. Weight bearing on the heel is recommended to minimize soft tissue trauma to the forefoot and promote healing of the incision. Return to play after surgical intervention for sesamoid dysfunction is typically between 2 and 4 months after surgery. When



Fig. 121.3 Plantar medial incision to tibial sesamoid.

shaving of the tibial sesamoid is performed, the athlete is allowed to resume running about 6 weeks after surgery.

Results

Sesamoidectomy has not been described as a treatment for acute fractures or nonfragmented nonunions of the sesamoids. However, it has been described for sesamoiditis. Saxena and Krisdakumtorn reported a mean average return to activity between 7.5 and 12 weeks in 26 patients.⁵⁵ Eleven professional/varsity athletes returned to sports at a mean of 7.5 weeks, whereas 13 “active” patients returned to sporting activity at a mean of 12 weeks.

Complications

Saxena and Krisdakumtorn reported two cases of hallux valgus, one case of hallux varus, and two cases of postoperative scarring with neuroma-type symptoms in their series, representing a 19% complication rate.⁵⁵ Transfer metatarsalgia may also occur.

METATARSALGIA

The term *metatarsalgia* is defined as pain of the plantar forefoot beneath the second, third, and fourth metatarsal heads.^{56,57} It is an umbrella diagnosis incorporating many etiologic origins. A key aspect in managing metatarsalgia is to assess the biomechanical factors involved.⁵⁸ Metatarsalgia can be subdivided into primary, secondary, and functional categories along with pathologic

conditions that arise during the stance and propulsive phases of gait.^{57,59} Primary metatarsalgia is due to anatomic abnormalities of the metatarsal lengths, tightness of the gastrocnemius, or hindfoot abnormality resulting in forefoot overload.⁵⁸ The first ray bears 40% of the weight distribution of the forefoot, with the lesser four rays sharing the remaining 60%.⁶⁰ Bearing in mind that the first, fourth, and fifth metatarsal-tarsal articulations allow considerable flexibility compared with the more rigid second and third metatarsal-tarsal articulations,⁶¹ any factor that lessens the ground force of the first metatarsal can increase the pressure applied to the lesser metatarsal heads. Metatarsal length discrepancy can similarly cause overload of one or more of the metatarsal heads. The most common pathology is the presence of an excessively long second metatarsal.⁶² Static metatarsalgia is often due to metatarsal slopes, whereas propulsive metatarsalgia is related to abnormal lengths.⁵⁸ During gait, the pressure center transfers under the metatarsal heads when the knee is extended and the ankle is in neutral dorsiflexion, between the 60th and 90th percentiles of the stance phase. In this critical phase, the knee is extended and gastrocnemius tightness may overload the forefoot, resulting in metatarsalgia.⁶³ Hyperextension of the lesser toe MTP joint during gait can produce a dynamic plantar protrusion of the metatarsal head as the toe is extending and the metatarsal head is forced plantarward.⁵⁷

An IPK is a localized callosity occurring on the plantar aspect of the foot.⁶⁴ A generalized callus can develop in the forefoot of an athlete as a result of increased pressure, and it is normal for a moderate amount of callus to form; in contrast, an IPK is a well-localized keratotic lesion. DiGiovanni et al. studied a population of patients presenting with forefoot pain and found the incidence of gastrocnemius contracture to be 65% (out of 34 patients).⁶⁵ Many authors have emphasized the importance of gastrocnemius tightness in producing forefoot symptoms and the positive effect on the gait pattern when such tightness has been overcome.^{66,67}

History

Barouk emphasized the distinction between stance and propulsive metatarsalgia and advocated different treatments for these conditions.⁵⁹ It is thus important to obtain a thorough history of the aggravating and relieving factors of the pain to try to discern between the two conditions. Athletes often report increased symptoms with specific activities or footwear. In persons with IPK, pain is usually directly under the callus.

Physical Examination

Physical examination first entails careful evaluation of the alignment of the foot when standing. The position of the hindfoot, the arch, and any great toe or lesser toe deformity should be noted. When a pes cavus deformity is suspected, a Coleman block test should be performed to determine if a fixed flexion deformity of the first metatarsal is present.⁶⁸ The plantar aspect of the foot should be carefully evaluated because the location of callosities can indicate the underlying problem. Stance-phase metatarsalgia generates proximal localized callosities or pain, whereas propulsive-phase metatarsalgia generates distal pain and a linear-type callosity.⁵⁹ Trimming of the callosity usually reveals

a well-circumscribed keratotic lesion. Solitary keratotic lesions are sometimes difficult to distinguish from a wart; however, keratotic lesions occur directly under a bony prominence, and unlike warts, they are avascular.

A discrete IPK caused by a prominent metatarsal head lateral condyle can be seen on the lateral side of the metatarsal head, typically under the fourth metatarsal. Identification of this IPK is important because it is amenable to simpler surgical correction with a quicker rehabilitation and recovery and is a more predictable procedure than is a metatarsal osteotomy.

With suspected stance-phase metatarsalgia, the foot should be evaluated for abnormal plantar flexion of the lesser metatarsals. Elevation of the first metatarsal pushes the entire load of the second rocker onto the second metatarsal, resulting in isolated keratosis underneath the second metatarsal head.⁵⁷ Propulsive keratosis, however, occurs during the third rocker of the gait cycle when external rotation of the lower limb results in a shearing force. This produces a more rounded appearance of the callosity, which extends distally toward the toe.⁵⁷ Hypermobility of the first ray should be evaluated along with midfoot mobility, because an unstable first ray can overload the lesser metatarsals.⁶⁹

Each MTP joint should be palpated to assess for position, synovitis, and joint stability because instability of the lesser toe MTP joints is a common but often under recognized cause of metatarsalgia.⁷⁰ The plantar flexion power of a lesser toe can be assessed using by the paper-pull test, where the patient tries to resist a strip of paper from being pulled from beneath the terminal phalanx. Inability to resist the paper being pulled from beneath is positive and may suggest a plantar plate tear.⁷⁰ The plantar plate provocation test can be used to differentiate plantar plate tears from Morton neuroma. In this test, the patient is seated with the foot hanging freely, and the examiner extends the toe, placing a plantar-directed force to the proximal phalanx to stress the plantar plate. If this causes pain, it is shown to correlate with subsequent findings of plantar plate injury on MRI.⁷¹ Pain at the second metatarsal head, a positive drawer test, and edema have high sensitivity, whereas previous first ray surgery or crossover toes have high specificity.⁷²

Each intermetatarsal web space should be palpated to assess for tenderness of the interdigital nerves. The Silfverskiöld test is performed to evaluate for contracture of the gastrocnemius and the gastrocnemius-soleus complex; the foot is maintained in an inverted position to avoid dorsiflexion movements at the midtarsal joints.^{57,73,74}

Imaging

Evaluation includes weight-bearing dorsoplantar and lateral views of the foot. Sesamoid views are included, depending on the location of the pain. The length of each metatarsal (MT) is assessed. Maestro et al. described the “harmonic parabola” where, in gait analysis studies, they considered the lateral sesamoid of the first MTP joint as a pivot.⁶² They also emphasized the longitudinal axis of MT2; the line passing through the center of the lateral sesamoid and bisecting a line perpendicular to the MT2 longitudinal axis is termed the Maestro line.⁵⁹ This line must pass through the center of the MT4 head to provide metatarsal harmony. Another method for assessing the relative lengths of

MT1 and MT2 was described by Hardy and Clapham, whereby a line is drawn along the axis of MT2 to bisect a transverse tarsal line drawn along the posterior articular surface of the cuboid. From this intersection, arcs are drawn at the level of the MT1 and MT2 articular surfaces, and the difference is assessed in millimeters. In an asymptomatic foot, MT1 has been shown to be between 2 and 4 mm longer than MT2 (mean 2 mm) using this method.⁷⁵ The kinetographic effects of a relative short MT1 with overload of the MT2 head and subsequent hypertrophic changes have been well described by Morton.⁷⁶ The slope of each MT and the difference in diaphyseal inclination between MT1 and MT2 are assessed on the lateral view. It is important to assess the oblique radiographs also to determine whether an apparently short metatarsal is truly short or whether it is apparently short due to increased plantar obliquity of the ray.⁷⁷ Radiographic evaluation may contribute to the diagnosis of a subluxated or dislocated MTP joint. Careful attention is given to bony abnormalities that may be responsible for areas of increased pressure. If the IPK is large, then ultrasonography can be helpful because dermoid inclusion cysts have been reported.⁷⁸

An extended sesamoid view of the metatarsal heads end-on with the toes dorsiflexed out of the way is useful in identifying a prominent metatarsal head lateral condyle and abnormalities in the transverse arch.

Decision-Making Principles

If a keratotic lesion continues to be symptomatic and significantly impairs athletic function, surgical intervention is considered. Because of the lengthy postoperative recovery time and the possibility of restricted MTP motion, recurrence, or the development of a transfer lesion, a rigorous trial of trimming, padding, and orthotic management should be carried out before surgery is performed. Surgical release of a gastrocnemius contracture must be considered with great caution in athletes because a gastrocnemius recession may result in the weakening of push-off. However, more recently, interest in performing a proximal medial gastrocnemius recession has increased. Abbassian et al. have demonstrated that medial head release at the knee can be performed safely with use of a local anesthetic followed by immediate loading, and most importantly, the power loss is minimal.⁷⁹

If all conservative efforts to treat metatarsalgia fail, then surgery is necessary. The goal of surgery is to distribute pressure evenly along the forefoot, thus reducing high-pressure areas. It is therefore essential to consider the clinical examination, pressure distribution, and x-ray evaluation to carefully plan a metatarsal osteotomy if this procedure is required. One should plan to correct metatarsal length to harmonize the parabola and elevate prominent heads to offload them.

Treatment Options

Nonoperative

The initial treatment of an IPK of any cause involves trimming the lesion to reduce the keratotic buildup. Trimming a callus also helps in differentiating it from a wart.

When the callus has been trimmed, a soft metatarsal pad is placed proximal to the keratosis to redistribute the pressure more uniformly. The use of a soft insole can alleviate the pressure



Fig. 121.4 After release of the gastrocnemius fascia, improvement is noted in the dorsiflexion range of motion of the ankle with the knee extended.

further in athletes.⁸⁰ Athletic footwear should provide a wide toe box and a soft sole to lessen impact when running. A Plastazote orthosis can be fabricated to relieve pressure beneath the IPK and provide correction for a postural deformity. Patients with tightness of the calf, be it gastrocnemius or the gastrocnemius-soleus complex, are taught stretching exercises to lengthen the muscles and thereby decrease pressure at the forefoot. Rocker-bottom shoes may also be helpful in reducing symptoms of metatarsalgia.⁵⁸

Operative

In all cases, the presence of a gastrocnemius contracture is treated with surgical release if the patient has persistent pain after engaging in a regimented nonoperative program (Fig. 121.4). In a series of 73 patients, some as young as 18 years of age (mean 50 years), gastrocnemius recession using Strayer technique was performed as an isolated primary procedure. All patients had positive Silfverskiöld test prior to surgery, and 38% ($n = 28$) had metatarsalgia preoperatively. Of these, only half reported excellent or good results, with visual analog scale pain scores decreasing from a mean of 5.6 preoperatively to 2.3 postoperatively at a mean follow up of 45 months.⁸¹ Fair evidence (grade B) supports gastrocnemius recession for the treatment of isolated foot pain due to midfoot-to-forefoot overload syndrome in adults.⁸² Mann and DuVries proposed that a small, discrete, IPK is caused by a prominent fibular condyle on the plantar aspect of the metatarsal head.⁸³ DuVries described a plantar condylectomy to correct this deformity.⁸⁴ Mann and Mann recommend removal of 20% to 30% of the condyle through a dorsal incision.⁶⁴ This procedure was later modified by Mann and DuVries, who performed an MTP arthroplasty, removing about 2 mm of the articular surface along with the plantar condylectomy.⁸³ After MTP joint arthroplasty, MTP joint motion is diminished by 25% to 50%. Although stiffness does not always affect function in a sedentary person, a competitive athlete typically requires more normal motion; if so, an MTP arthroplasty is contraindicated.



Fig. 121.5 Intraoperative appearance of the metatarsal head after fixation with a 2.4-mm screw. Note the dorsal overhanging bone that must be removed to allow for appropriate articulation. (From Seybold J, Kadakia AR. Claw and hammer toes. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

Elevation of the metatarsal head can be achieved with use of a vertical chevron procedure.^{83,85,86} A distal oblique osteotomy,⁸⁷⁻⁹² capital oblique (Weil) osteotomy,^{57,59,93,94} and three-step (Maceira) osteotomy^{57,95} are designed to both shorten the metatarsal and elevate the metatarsal head (Fig. 121.5). These procedures are best used in the treatment of propulsive-phase metatarsalgia. It is believed that elevation of about 3 mm is necessary for adequate decrease of pressure beneath the symptomatic metatarsal head.⁹⁶ Hansen described midshaft segmental osteotomy in which two parallel cuts are made in the bone, which is subsequently fixed with a four-hole plate.⁹⁷ Galluch et al. reported good results with this osteotomy in 95 patients. A 99.2% union rate was reported, and transfer lesions developed in 5 patients.⁹⁸

Postoperative Management

A compressive dressing is applied at the time of surgery, and the patient is allowed to bear weight in a postoperative shoe. After a metatarsal osteotomy, adequate time must be allowed for healing. Premature athletic activity may lead to failure of fixation, displacement of an osteotomy, or nonunion. In general proximal metatarsal osteotomies require 6 to 12 weeks for osseous union to occur. A vertical chevron osteotomy performed with a thin oscillating saw blade heals in about 6 weeks. After an MTP joint arthroplasty and condylectomy, about 6 weeks is necessary for adequate healing to occur at the MTP joint.

Once adequate healing has occurred (generally in 4 to 6 weeks), gentle ROM is initiated at the MTP joint to diminish postoperative stiffness. When radiographs demonstrate bony union, aggressive walking activity is initiated. Taping of the forefoot or the use of soft metatarsal pads helps to alleviate symptoms with athletic activity during recovery from surgery. About 4 weeks after the initiation of walking, jogging is initiated, followed by running activities as pain permits.

Authors' Preferred Method

Metatarsalgia

When a competitive athlete requires a more normal motion, a distal metatarsal osteotomy is generally considered the osteotomy of choice. A number of osteotomies have been described in both treatment of stance-phase and propulsive-phase metatarsalgia, although distal osteotomies are generally used to treat propulsive metatarsalgia.

The distal metatarsal is approached dorsally, between the metatarsals if two are to be operated on. The collateral ligaments are released in order to release the plantar tissues in case of a hammer-toe deformity; if the toe is subluxed/dislocated, the plantar plate is likely to be torn and will require repair. The collateral ligaments may require reefing in case of coronal plane deformity. The most important step in a distal metatarsal osteotomy is to make the cut as parallel to the weight-bearing axis of the foot as possible to ensure that shortening does result in plantar flexion of the metatarsal head. Next, a parallel slice of the metatarsal is removed according to the amount of shortening/elevation that is required. This osteotomy is fixed with a screw and is performed under image intensifier control to assess the metatarsal parabola (Fig. 121.6).



Fig. 121.6 An anteroposterior radiograph of a patient after Weil osteotomy of the second and third toes. The normal cascade of the foot is preserved, with the third toe slightly shorter than the second. Maintaining this cascade is critical to prevent iatrogenic metatarsalgia. (From Seybold J, Kadakia AR. Claw and hammer toes. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

Decreased ROM is a significant risk after MTP arthroplasty as well as after a distal metatarsal osteotomy. In the postoperative follow-up, one should pay close attention to any impending deformity or stiffness; it will be easier to start passive stretching or manipulation with use of a block or anesthetic earlier rather than later. In athletes, plantarflexion is very important. Full return to athletic activity is expected when bony healing occurs. Usually between 6 and 12 weeks after surgery, aggressive walking is

initiated. Walking is advanced to jogging after 4 weeks as permitted by pain and swelling. The athlete then increases running activities as tolerated, gradually returning to sport between 3 and 6 months after surgery.

Results

In the treatment of an IPK, Winson and coworkers found that 53% of their patients had significant postoperative symptoms, including transfer lesions in 32%, nonunion in 13%, and an overall recurrence of an IPK in 50% of patients.⁹² In 66 of the 124 feet, major postoperative complaints were noted. The authors stressed that in either a cavus or a rigid foot, a distal oblique sliding osteotomy was contraindicated.

In the treatment of propulsive stance metatarsalgia, the Weil-type or triple-step osteotomy is becoming the most accepted practice. Good to excellent long-term results have been noted in 70% to 100% of patients treated with a Weil-type osteotomy.⁹⁹⁻¹⁰¹ The three-step osteotomy was introduced to try to reduce complications observed with a Weil osteotomy. Espinosa et al. reported that joint stiffness and floating-toe deformities were rarely seen with this type of osteotomy.⁵⁷ Pérez-Muñoz et al. reported on 82 patients treated with a Weil or three-step osteotomy; they found recurrence of metatarsalgia in 4.3%, moderate joint stiffness in 60.2%, floating toes in 4.3%, and delays in bone healing in 7.5%. All delayed unions occurred in the triple osteotomy group and were presumed to be a result of the decreased stability of the osteotomy.¹⁰²

Mann and DuVries evaluated 100 patients with discrete IPKs and noted a recurrence rate of 17.6% after MTP arthroplasty. Of these recurrences, 5% were found under the symptomatic metatarsal. A transfer lesion developed in 13% of cases. Despite these results, 93% of the patients were satisfied with their outcome.⁸³

Complications

Complications reported with a Weil-type osteotomy include joint stiffness, floating-toe deformity, and transfer metatarsalgia. Stiffness may be caused by morphologic alterations of the MTP joint postoperatively, fibrosis, reaction to osteosynthesis material, or biomechanical alterations of the intrinsic musculature.^{57,99,103} Beech et al. noted that postoperative joint ROM was reduced in most cases.⁹⁴ Floating-toe deformity has been reported in up to 30% of cases.¹⁰⁴

A 5% rate of complications was found after condylectomy and MTP arthroplasty for discrete IPK, which included fracture of the metatarsal head, avascular necrosis, drift of the involved toe, and cock-up of the involved toe.⁸³

Future Considerations

With further study of the effect of a gastrocnemius contracture on the incidence of forefoot pain, a shift away from bony correction of the foot pathology may occur if the outcomes are favorable. The morbidity from forefoot surgery as described is a major detriment to performing these operations in the athlete, and the possibility of a soft tissue correction without subsequent loss of athletic function would be very desirable.

HALLUX RIGIDUS

History

Typical presenting symptoms of hallux rigidus are pain overlying the MTP joint from cartilage degeneration possibly combined with mechanical shoe pain as a result of a dorsal bony prominence. The pain worsens over time with activity and typically improves with rest and elimination of restrictive footwear. Athletes will report a reduction in motion when crouching or during push-off. Once pain at night or at rest is noticed, the degeneration within the joint is generally widespread. Plantar pain suggests degeneration in the inferior surfaces of the MTP joint or the MT-S joints and is a relative contraindication for cheilectomy.^{105,106}

Examination

Hallux rigidus is a restriction of motion due to degenerative changes in the first MTP joint. The MTP joint may demonstrate a plantarflexed position with a dorsal prominence (Fig. 121.7). Tenderness typically starts dorsolaterally on the joint line and is worse at the extremes of motion. With increased severity, the patient may have pain with the “grind test,” in which the examiner impacts the phalanx on the first metatarsal head in the central ROM. A positive test is consistent with end-stage arthritis, denoting that a simple cheilectomy will not be effective. Compression of the first MTP joint during assessment of ROM can help the clinician decide whether the articular surface glides freely and indicate intact cartilage.¹⁰⁶

Hallux rigidus should be considered clinically as either functional or structural, and it is essential to examine the ROM at the first MTP joint during both non-weight bearing and weight bearing to assess the influence of gastrocnemius tightness. Patients who have good passive dorsiflexion range when non-weight bearing but a reduction during weight bearing are considered to have functional hallux rigidus. In these patients osteotomies should enable a change in functional arc of motion and improve symptoms. Functional hallux rigidus is assessed with the patient supine and upward pressure placed on the first metatarsal head

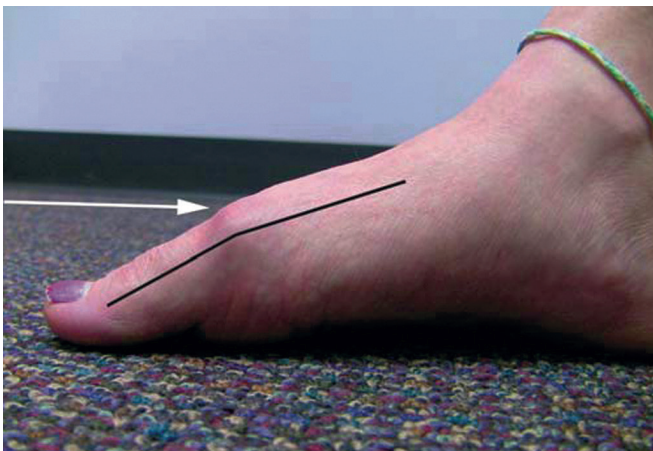


Fig. 121.7 A patient with hallux rigidus with a fixed flexion deformity of the hallux. Note the large dorsal prominence (arrow). (From Seybold J, Sabb B, Kadakia AR. *Arthritides of the foot and ankle*. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

to simulate weight bearing and bring the ankle to neutral position. The first MTP joint should have at least 60 degrees dorsiflexion in this position. When dorsiflexion is applied to the first MTP joint in this position, the movement should occur at the first MTP joint. If the ankle plantarflexes to achieve the ROM at the MTP joint, this is considered functional hallux rigidus.¹⁰⁷ If patients have restricted ROM during both non-weight bearing and weight bearing (i.e., structural hallux rigidus), arthrodesis or arthroplasty is preferred.¹⁰⁸

Imaging

Radiologic classification systems have been proposed but do not correlate well with clinical features.¹⁰⁹ Narrowing of the joint, dorsal spurs from the metatarsal and phalanx, and loose body formation are common findings.

Decision-Making Principles

Although the clinical and radiographic severity of hallux rigidus can be classified according to Coughlin and Shurnas, ranging from stiffness with normal radiograph to stiffness and pain throughout the first MTP joint range with more than 50% joint space narrowing and severe dorsal osteophytes, it must be remembered that the natural history of hallux rigidus is inconsistent.^{109,110} Conservative measures must always be tried in the first instance for all athletes; thus the real decision-making lies in the choice between joint-sparing and joint-loss procedures. Clearly sporting requirements must be considered, but some key clinical features will aid in the decision-making. For instance, pain at night or at rest, crepitation upon passive motion, or a positive grind test suggests a widespread problem that will not respond to a joint-sparing option and instead requires either fusion or a motion-sparing salvage procedure. As long as patients understand that outcomes may be reduced, cheilectomy can be considered even in the presence of radiographic degeneration provided that there is no significant bone loss.¹⁰⁶

Similarly, sesamoid pain will not be alleviated by an MTP cheilectomy. On the other hand, a smooth arc of motion (albeit reduced) and pain at the end of the range suggest that a cheilectomy is feasible.

The issue becomes difficult in cases of a failed cheilectomy or when the choice between cheilectomy and fusion is not clear-cut; in these cases, radiographic findings and visualization of the joint (arthroscopy is better than MRI) is helpful in determining whether the joint is salvageable.

In patients with symptomatic functional hallux rigidus, distally based metatarsal osteotomies such as the Watermann-Green, Youngswick, and oblique distal metatarsal osteotomy are useful. All of these options result in shortening of the MT1 and therefore are best performed in cases of an elongated first ray, where shortening is desirable. The Watermann-Green is versatile and enables rotation of the distal bone to redirect intact plantar cartilage to a more dorsal position. However, it has been suggested that if this is not done with caution, symptoms may occur at the MT-S joint due to increased contact pressures. The Youngswick osteotomy is similar but the plantar limb is at 60 degrees rather than 135 degrees to the dorsal cut and is more reproducible, resulting in shortening and depression of the metatarsal head. Both

osteotomies are inherently stable. Another option is the oblique distal osteotomy, which is at 30 degrees to the sagittal plane and allows shortening and depression of the metatarsal head but has no scope for realignment of the articular surface. For this reason it is best used if the articular surface does not need realignment. In practice, all of these procedures are combined with a cheilectomy.¹⁰⁸ When there is functional hallux rigidus with an elevated first ray but no arthritis of the first MTP joint, a proximal plantarflexion osteotomy has been advocated.¹⁰⁸ Clinical heterogeneity of outcome studies with small numbers of study participants makes it impossible to draw definitive conclusions as to which osteotomy is superior.¹¹¹

Treatment Options

Nonoperative

Surgery is not advised for asymptomatic functional hallux rigidus.¹⁰⁷ Certain activities such as dancing or sprinting demand a high degree of dorsiflexion. These cases are difficult to manage conservatively other than to treat the patient with intra-articular corticosteroid injections and nonsteroidal anti-inflammatory drugs. However, in a recent questionnaire study of patients who underwent corticosteroid injection for hallux rigidus, although 20 of 22 (91%) of patients reported a benefit, only 3 (14%) reported benefit that lasted more than 3 months. In view of this, corticosteroids have a limited role in management of hallux rigidus.¹¹²

Because injections are likely to accelerate the wear within the joint, they must be used judiciously. The mainstay of conservative treatment is stabilization of the first MTP. Taping and orthotics can be very useful in pain control, but these measures, of course, reduce the joint's motion.¹⁰⁹ A double-blind clinical trial of patients with hallux limitus, comparing custom-made in-shoe orthotics that elevated the first MTP joint versus the same orthotic without elevation, found a significant increase in maximal dorsiflexion in the intervention group of 22.2 degrees as compared with the control group, which increased range by 5.65 degrees. However, this study excluded patients with first MTP joint pain or radiographic osteoarthritis.¹¹³

Operative

Joint-sparing procedures should be performed for athletes if at all possible. If such procedures are not appropriate, joint-destructive but motion-sparing procedures (e.g., an osteotomy or interposition arthroplasty) should be considered. However, metallic or Silastic joint replacement has no role in the athlete. Ultimately fusion surgery is a durable and long-term solution that allows robust activity.

Cheilectomy. A cheilectomy is the first line in surgical management. Impingement always starts dorsally and spreads downward except in traumatic lesions, which start centrally; thus a cheilectomy is not appropriate for this group. Removal of the dorsal 25% of the joint and the osteophyte is very successful at removing the footwear irritation and the end-of-range impingement pain. It does not usually improve the ROM but prolongs the athlete's career, even though it does not slow down the rate of progression (Fig. 121.8). In a recent single-surgeon study of 179 patients who underwent open cheilectomy, 58 patients were

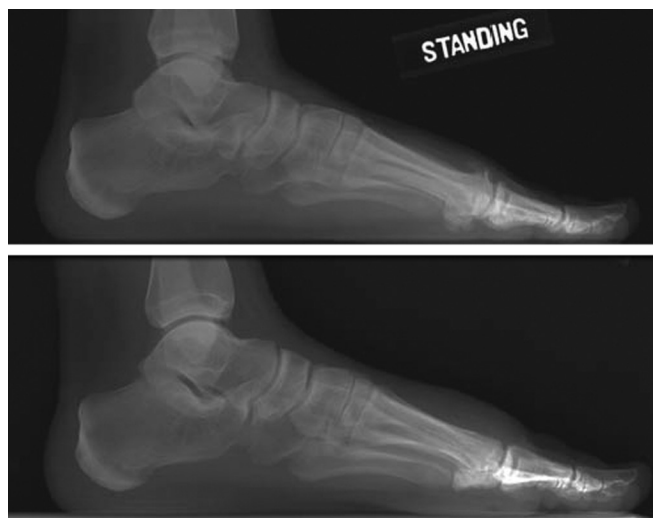


Fig. 121.8 Preoperative and postoperative radiographs of a patient with grade II hallux rigidus who underwent a cheilectomy. The postoperative radiograph demonstrates removal of the dorsal osteophytes from both the metatarsal and the phalanx. The final appearance of the metatarsal reveals colinearity, with the metatarsal shaft indicating adequate bony resection. (From Seybold J, Sabb B, Kadakia AR. Arthritides of the foot and ankle. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

followed up at a mean follow up of 7.14 years (33.5% response rate). The mean percentage improvement in pain score was 87.71%, with 88% returning to normal function, including exercise. Overall, 95% of patients stated that they would undergo the procedure again whereas 88% reported either no pain or occasional pain.¹¹⁴ Approximately 3% of patients required subsequent first MTP joint arthrodesis.¹¹⁴

Arthroscopic treatment of hallux rigidus. Arthroscopic cheilectomy is preferred when the articular surfaces require direct inspection, for example when the radiographic findings do not correlate with symptom severity.¹⁰⁶ Fusion is also possible arthroscopically¹¹⁵ and consent should include the intraoperative decision to convert to fusion if the joint has severe degeneration. For arthroscopic cheilectomy, the osteophyte can be marked with Kirschner wires under fluoroscopic imaging to penetrate the articular surface; then resection is completed arthroscopically with reference to these.¹¹⁵ Other authors do not use wires to mark resection for arthroscopic cheilectomy but resect until 50 to 70 degrees of passive dorsiflexion is achieved.¹⁰⁶

Minimally invasive treatment of hallux rigidus. Minimally invasive first MTP joint fusion, using high-speed burr under radiographic imaging, enables patients to mobilize full weight bearing postoperatively, with 90% achieving clinical union at 6 weeks of follow-up and 85% having radiographic evidence of fusion at 1 year.¹¹⁶ Minimally invasive cheilectomy results in significant reductions in pain at a minimum follow-up of 1 year.¹⁰⁵

Osteotomy. Moberg osteotomy is used in young active patients with moderate to severe hallux rigidus (Hattrup and Johnson grades II and III), to augment cheilectomy where conservative treatment has failed.¹¹⁷ The Moberg procedure dorsiflexes the proximal phalanx, enabling greater movement through the third rocker before the MTP joint must engage; thus it results in a



Fig. 121.9 A first metatarsophalangeal arthrodesis using a dorsal plate and lag screw for fixation. (From Seybold J, Sabb B, Kadakia AR. Arthritides of the foot and ankle. In: Miller MD, Sanders TG, eds. *Presentation, Imaging and Treatment of Common Musculoskeletal Conditions*. Philadelphia: Elsevier; 2011.)

functional increase in the dorsiflexion of the joint and can also decompress the joint.¹¹⁸ A dorsal closing wedge is performed in combination with a cheilectomy; it is very stable and heals rapidly. If the joint has poor plantarflexion, the toe may be too elevated, causing problems with footwear in stance. Rehabilitation is similar to that for a cheilectomy. An important preoperative consideration for performing the Moberg procedure is to determine whether the patient has adequate first MTP joint plantarflexion preoperatively, which will be lost after the procedure. The patient should be counseled for this and also for the intraoperative decision to proceed with arthrodesis if the joint is found to have severe arthritis.¹¹⁷

Two metatarsal osteotomies can be considered, especially in the athlete with advanced degeneration who runs. In the long or elevated metatarsal, a shortening/plantarflexing osteotomy (the chevron type is the most stable) can be performed and is straightforward to plan, execute, and fix. If the metatarsal length is normal, then a dorsally rotating distal oblique osteotomy can be performed.

Motion-sparing procedures in advanced degeneration. Resection or interposition arthroplasty, although not joint sparing, can provide excellent pain relief while maintaining motion and may be preferable to arthrodesis in certain athletes for whom dorsiflexion is important, such as dancers or runners. Cosmesis may be poor with a short, elevated toe, but the range of movement may be excellent. Recently synthetic cartilage hemiarthroplasty of the first MTP joint using polyvinyl alcohol (PVA) hydrogel was found to result in functional outcomes and pain relief comparable to first MTP joint fusion, with the benefit of maintaining motion. However, the long-term outcomes and results in athletes are currently not known.^{119–121} Resection or Keller arthroplasty is not recommended in a young active population, although the Valenti procedure—which resects the dorsal metatarsal head and dorsal phalangeal base and maintains length and plantar loading—can obviate some of these issues.¹²²

Metatarsophalangeal arthrodesis. MTP arthrodesis remains the gold standard of pain relief and function.¹²³ Although it comes at the cost of sacrificing any remaining movement, generally very little movement is left in persons with late-stage disease

anyway. However, in athletes who require dorsiflexion or who have a short great toe, it should be used with caution. Care should also be taken in the person who has significant motion of the interphalangeal joint with uncovering of the phalangeal head and loading of the head. Loss of motion at the MTP joint will increase the loading on the phalangeal head in these cases, and this situation can be very difficult to manage.

The joint should be fused in 10 to 15 degrees of dorsiflexion relative the weight-bearing position of the foot to maximize the third rocker without setting the toe too high, in which case it may rub on the toe box (Fig. 121.9). We assess this simulated weight-bearing position intraoperatively with a footplate and ensure that the pulp of the hallux is approximately 1 cm off the plate. This position may also be achieved by placing a fingertip under the hallux pulp in the simulated weight-bearing position. This has been shown to equate a first MTP joint extension angle of between 1 and 10 degrees relative to the ground and is well tolerated clinically.¹²⁴ Fusion angles of less than this may place the interphalangeal joint at risk of subsequent degenerative change. Use of normal footwear can be resumed at 6 weeks after weight bearing on the heel in a protective shoe from the time of surgery. Participation in sports can generally be resumed 3 to 4 months after surgery. Nonunion mandates further surgery, but the rates with current implant techniques are very low.

Results

In appropriate cases, cheilectomy has excellent outcomes, and rehabilitation is rapid. Approximately 70% of patients obtain excellent relief, 10% obtain modest relief, and 20% get no relief.¹²⁶ Microfracture of lesions below the level of the osteotomy or of isolated central lesions has an unreliable outcome.

Percutaneous and arthroscopic techniques appear to improve the rate of rehabilitation, but few data are as yet available on this subject, with no studies comparing open versus arthroscopic treatment of hallux rigidus.¹¹⁵ Arthroscopy allows visualization of the joint and therefore may be preferable. It is generally at least 8 weeks before running sports can be commenced, although cycling and swimming can be commenced earlier. It takes 3 to 4 months to achieve maximum outcome.



Authors' Preferred Technique

Hallux Rigidus

Cheilectomy is the mainstay of treatment of the athlete. The joint is approached dorsally and the collaterals are released to allow full visualization; it is important to free up the sesamoids and plantar plate because they are frequently adherent to the underside and can restrict movement. After removal of the dorsal 25% of the head, it is refashioned to make it rounder and more congruent. One should attempt to achieve at least 80 degrees of dorsiflexion; if this degree of dorsiflexion is not achieved, a dorsiflexing osteotomy of the proximal phalanx (a Moberg osteotomy) may be considered. Patients are able to bear weight and start gentle active motion immediately and can begin full rehabilitation once the wound has healed.

MTP fusion remains the gold standard in terms of pain relief and longevity, but if movement is important, our preferred motion-sparing option is the interposition arthroplasty.¹²⁵ A cushion of soft tissue is made from periosteum, attached fat, and tendon and stabilized by suturing to the plantar capsule. Alternatively, regenerative tissue matrix may be used as the interposition material. Implant arthroplasty has no role in the athletic population.

Complications

A cheilectomy can work very well, even in persons with radiologically advanced disease. However, a significant number of patients will have ongoing pain and stiffness that can limit participation in sports, and a small number may have significant arthrofibrosis that can prevent return to sport. This group is best served with revision to a fusion, although a course of intra-articular steroids may be of some benefit.

With modern fixation techniques, outcomes from fusion are reliable. Shortening of the big toe can cause rubbing on the end of the second toe, but this problem can readily be treated. A more difficult issue is the case with painful loading on the proximal phalangeal head, which occasionally requires conversion of a fused MTP joint into an interposition arthroplasty.

For a complete list of references, go to ExpertConsult.com.

SELECTED READINGS

Citation:

Anderson RB. Turf toe injuries of the hallux metatarsophalangeal joint. *Tech Foot Ankle Surg.* 2002;1(2):102–111.

Level of Evidence:

III

Summary:

The author provides an excellent review of turf toe injuries combined with an algorithm of treatment from the surgeon's extensive experience.

Citation:

Hong CC, Pearce CJ, Ballal MS, et al. Management of sports injuries of the foot and ankle: an update. *Bone Joint J.* 2016;98-B(10):1299–1311.

Level of Evidence:

III

Summary:

The authors provide an excellent contemporary review of turf toe injuries.

Citation:

Kadakia AR, Molloy A. Current concepts review: traumatic disorders of the first metatarsophalangeal joint and sesamoid complex. *Foot Ankle Int.* 2011;32(8):834–839.

Level of Evidence:

II

Summary:

The authors provide a comprehensive review regarding the diagnosis and treatment of injuries to the metatarsophalangeal (MTP) joint and sesamoids. The review covers the current level of evidence and grades of recommendation for the treatment of first MTP capsuloligamentous injury and sesamoid pathology.

Citation:

Cychosz CC, Phisitkul P, Belatti DA, et al. Gastrocnemius recession for foot and ankle conditions in adults: evidence-based recommendations. *Foot Ankle Surg.* 2015;21(2):77–85. <http://doi.org/10.1016/j.fas.2015.02.001>.

Level of Evidence:

IV

Summary:

In this topical review, the authors provide an up-to-date summary of the evidence for gastrocnemius recession.

Citation:

Mann RA, Mann JA. Keratotic disorders of the plantar skin. *J Bone Joint Surg Am.* 2003;85(5):938–955.

Level of Evidence:

III

Summary:

The authors of this excellent review cover the possible etiologies and discuss how to address the specific pathologic condition accordingly.

Citation:

Seibert NR, Kadakia AR. Surgical management of hallux rigidus: cheilectomy and osteotomy (phalanx and metatarsal). *Foot Ankle Clin.* 2009;14(1):9–22.

Level of Evidence:

III

Summary:

The authors provide a comprehensive review regarding hallux rigidus, with a focus on joint-sparing procedures. The technique and results of multiple operations are discussed.

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